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Official Statistics in the era of AI

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Abstract

Citizens, policymakers, and researchers expect more than ever from official statistics: timely, reliable products; detailed insights; and easier access to information. IstatData, the national data portal, plays a key role in this strategy. While quality and trust remain at the heart of Istat's mission, we are modernising dissemination to make official statistics more accessible, transparent, and user-friendly.

This transition poses significant challenges: protecting privacy, navigating complex regulations, ensuring transparency, and adapting quality standards to support user-friendly and machine-readable dissemination formats. IstatData provides users with direct access to official statistics, offering standardised formats, interactive tools, and rich metadata. New AI-based features include a tool that enables natural-language dataset search and another that creates real-time analyses and visualisations.

The journey is complex but fascinating. To ensure relevance and reliability, we must keep investing in research, skills, and modern tools—so that official statistics remain accessible and trusted by all users.

I. Introduction

Over the past decade, the environment in which National Statistical Institutes (NSIs) operate has undergone a significant transformation. The digitization of society, the widespread use of online services, and rapid technological advances driven by artificial intelligence (AI) have reshaped how citizens, policymakers, businesses, and researchers produce, access, and use information. As data ecosystems become more complex and dynamic, expectations for official statistics continue to grow: users increasingly demand timely insights, highly granular data, transparent methods, and simpler access to statistical information. Traditional dissemination models—mainly based on static tables, keyword searches, and metadata-driven navigation—are no longer sufficient to meet these evolving needs.

In this context, NSIs face a dual challenge. They must maintain the core principles that underpin trust in official statistics: methodological soundness, rigorous quality frameworks, transparency, and respect for privacy and confidentiality. At the same time, they need to embrace innovation by integrating AI and advanced data-science techniques into the design, production, and dissemination of statistical information. Balancing these two demands—innovation and trust—requires a strategic effort involving new skills, modern infrastructures, and governance frameworks capable of guiding responsible and sustainable technological adoption [1, 2].

At Istat, this transformation is evident in two key areas. First, in the integration of AI into statistical production, where machine-learning techniques and Large Language Models (LLMs) [3] are increasingly used to classify economic activities and products, extract information from non-traditional data sources such as balance sheets and patents, and support editing and imputation processes. These innovations aim to improve efficiency, reduce manual burden, and expand the analytical capabilities of official statistics.

Second, AI is reshaping data dissemination and user interaction. A prominent example is the evolution of IstatData, the Institute's data dissemination portal. In addition to providing access to thousands of multidimensional tables, the platform is becoming increasingly intelligent, offering natural-language searches, personalized table creation, interactive visualizations, and AI-powered reporting tools. These developments are part of a broader effort to make official statistics more accessible to non-experts and better aligned with the communication needs of an increasingly digital society.

The shift toward user-centred dissemination systems is motivated by a simple insight: statistical information must be not only accurate and reliable but also discoverable, interpretable, and usable. Modern AI techniques—such as text embeddings [4], semantic search [5], and retrieval-augmented generation (RAG) [6]—help bridge the gap between the technical language of official statistics and the natural language in which users express their information needs. By enabling queries such as "enterprises performing e-commerce" or "drinking habits of the Italian population," semantic search tools help users navigate statistical content more effectively, lowering barriers and opening new possibilities for data-driven knowledge.

However, this transformation also brings new risks and responsibilities. Integrating AI into statistical processes requires careful consideration of privacy, bias, transparency, and explainability. It also raises questions regarding the validation of black-box models, the sustainability of computational infrastructures, and the long-term interoperability of AI-generated outputs. These issues are central to the trust contract between NSIs and society. As such, they must be addressed with methodological quality, robust governance, and continuous investment in staff training and institutional capacity [2].

The goal of this paper is to describe Istat's ongoing journey toward AI-enabled statistical production and dissemination, with a particular focus on IstatData as a key platform for user-centred access to official statistics. Drawing on national partnerships, international projects, and ongoing experiments within the European Statistical System, we illustrate how AI is being incorporated into real workflows, what challenges remain, and how the principles of transparency, authenticity, and traceability can guide the responsible adoption of new technologies.

II. Challenges – Balancing Innovation and Trust in AI-Driven Statistics

The integration of artificial intelligence into official statistics offers unprecedented opportunities but also introduces complex challenges that statistical agencies must address to ensure responsible, trustworthy, and sustainable innovation. While AI can boost efficiency, support new data sources, and enhance user access to statistical information, it also brings methodological, technological, ethical, and organizational risks that cannot be ignored. For NSIs, the ability to balance innovation with trust is not merely desirable—it is fundamental to maintaining their role as authoritative producers of public information [1,7].

IIA. Protecting Privacy and Ensuring Data Protection

NSIs operate under strict legal frameworks that protect personal data, confidentiality, and the appropriate use of information. As AI technologies—particularly machine learning and Large Language Models (LLMs)—become more widespread, new risks emerge, including data leakage, unintended memorization, and the misuse of copyrighted or sensitive material. Ensuring compliance with privacy regulations requires careful design of data flows, robust governance for cloud-based solutions, privacy-enhancing techniques, and transparent procedures for managing training data. Approaches such as differential privacy, secure multi-party computation, and synthetic data generation are increasingly relevant, but they must be adapted to the statistical domain to preserve usability and analytical accuracy [8].

IIB. Safeguarding the Reputation of Official Statistics

AI systems can generate results that are difficult to interpret, verify, or replicate—particularly when models behave as black boxes. This raises the risk of errors, hallucinations, biases, or discrimination when algorithms are used to support policy or guide analytical processes. For NSIs, maintaining credibility requires establishing clear validation procedures, documenting model behaviour, and ensuring that AI-assisted outputs remain traceable, explainable, and aligned with quality standards. The reputational cost of incorrect or non-transparent results is particularly high in public statistics, where trust is a foundational asset [8].

IIC. Technological Constraints and Infrastructure Requirements

Many statistical offices still rely on traditional data centres designed for classical IT systems rather than for AI workloads. AI methods—especially deep learning and LLM-based approaches—demand high-performance computing, GPU clusters, and scalable infrastructures. This creates a strategic dilemma: **Should NSIs invest in upgrading on-premises infrastructure, or adopt cloud-based solutions?**

Both options involve complex trade-offs concerning cost, efficiency, data protection, regulatory compliance, sustainability, and long-term flexibility. Decisions about infrastructure will influence not only technical capabilities but also the scope and pace of future AI adoption.

IID. Methodological Challenges

The growing use of AI methods and the increasing reliance on new data sources require substantial revisions to existing methodological frameworks and quality standards. Traditional criteria were developed for structured, well-curated data and deterministic processes. In contrast, AI approaches often rely on probabilistic black-box models, heterogeneous data, iterative training, and feedback loops. This evolving landscape presents several methodological challenges:

- Defining validation procedures for black-box models.
- Developing metrics for assessing algorithmic stability and robustness.
- Adopting advanced privacy-enhancing techniques without compromising analytical value.

Modernizing quality standards is essential to ensure that AI-driven processes produce results that remain reliable, comparable, and aligned with the principles of official statistics.

IIE. Governance, Skills, and Organizational Readiness

AI adoption requires strong institutional governance that ensures coherence, accountability, and responsible innovation. At Istat, governance structures for AI are organised around four complementary domains:

1. **Innovation & Research**: focusing on experimentation, strategic identification of use cases, and development of AI solutions.
2. **Data & Rules**: which define policies, regulatory compliance, and data-governance responsibilities.
3. **People**: staff training, literacy programs, cultural change, and the evolving role of public employees.
4. **Partnerships & Collaborations**, which leverage public and private research bodies, academia, other NSIs, and international projects to support knowledge exchange and capacity-building.

Clear roles, responsibilities, and decision-making processes are essential to ensure that experimentation aligns with legal, ethical, and strategic objectives. Equally important is the development of new skills. Data scientists, methodologists, domain experts, IT specialists, and communication professionals must collaborate within multidisciplinary teams. Staff training and literacy programs are crucial to building internal capacity—not only to use AI tools, but also to understand their limitations and ensure their responsible adoption

III. AI Applications at Istat: Harnessing Non-Traditional Data Sources

Modern AI methods—such as text mining, semantic search, and Large Language Models (LLMs) [2-6]—are creating new opportunities for NSIs to integrate complex, unstructured, non-traditional data sources into their statistical processes. These data sources, which are increasingly important, are often rich in information but difficult to analyze using traditional methods. At Istat, AI techniques are gradually being integrated into various statistical processes, ranging from data acquisition to data dissemination. Key examples include:

- **Textual data sources** —such as balance sheets, patent abstracts, or medical notes—can be automatically classified using modern AI approaches. Sentence-embedding models enable semantic search algorithms to assign the correct code of an official classification (e.g., ATECO, ISCO) to each document, turning free text into usable statistical information. For instance, mapping patent abstracts to ISCO categories allows the estimation of occupational exposure to AI technologies.
- **Enterprise websites**, which often contain valuable information, such as product descriptions or information about firms' online activities. Through web scraping and modern AI techniques, it is possible to estimate characteristics such as the presence of e-commerce services or firms' engagement in social media activities. A current Istat project within the PRODCOM survey is testing the use of semantic search to infer the products manufactured by enterprises directly from textual content available on their websites.

These data sources represent both a challenge and an opportunity. Their heterogeneity, variability, and linguistic complexity make them difficult to process using traditional rule-based approaches. However, modern AI models—especially embedding-based semantic matching—enable more flexible and scalable solutions. By converting text into dense vector representations that capture semantic meaning, these models enable NSIs to classify, cluster, and extract structured information from unstructured content, thereby expanding the analytical scope of official statistics.

Building on these experiences, the following section introduces the basic idea behind semantic search and explains how it supports automatic coding and easier access to statistical data.

IV. Semantic Search: A Methodological Overview

Traditional information-retrieval systems in official statistics rely on keyword matching. This approach requires users to know the exact terms appearing in table titles, metadata fields, or classification labels—making relevant content difficult to find for non-experts or for users unfamiliar with statistical terminology.

Semantic search offers a more adaptable approach by focusing on the *meaning* of text rather than keyword matching. The fundamental idea is straightforward: both documents and user queries are converted into vectors using sentence-embedding models, and their similarity is calculated within this shared semantic space.

Semantic search workflows typically follow **two main phases**:

(1) *knowledge base creation*
(2) *matching user queries*.

1. Knowledge Base Creation

Before a semantic search system can process queries, all items that may be retrieved—such as classification codes, table metadata, or other text units—must be represented consistently. This requires:

- **Defining descriptors**, which summarise the relevant textual information associated with each item. Descriptors may include table titles, descriptions, keywords, or classification labels.
- **Encoding the descriptors** using a pre-trained sentence-embedding model. The model converts each descriptor into a dense vector capturing its semantic content.
- **Building the embedding knowledge base**, where all vectors are stored and indexed. This database serves as the base for the semantic search engine.

This step ensures that all elements of interest—whether ATECO or ISCO codes, IstatData tables—are embedded in the same semantic space.

2. Matching User Queries

Once the knowledge base has been generated, the system can process user queries:

- The **user query**, written in natural language, is embedded using the same sentence-embedding model.
- A **similarity function**—typically cosine similarity—compares the query vector with the vectors stored in the knowledge base.
- The system retrieves the items with the highest similarity scores, even when the query and the descriptors use different wording.

In this context, semantic search enables flexible matching between free-text queries and structured statistical content (see Figure 1). A query such as *"companies selling online"* can correctly retrieve tables related to e-commerce activities, even if the term *"selling online"* does not appear explicitly in the metadata.
Similarly, when applied to statistical production processes, semantic search can support **automatic coding**, in which free-text descriptions (e.g., balance sheets, patent abstracts, or respondents' free-text answers in questionnaires) are mapped to the most appropriate category in an official classification.

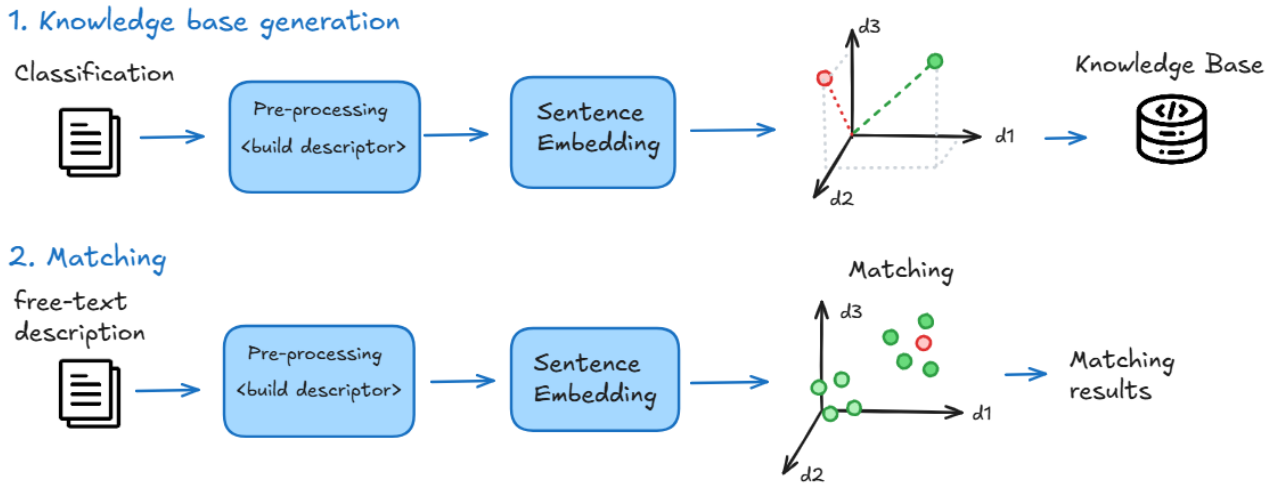

Figure 1: Overview of the semantic search workflow. The knowledge base is created by embedding textual descriptors associated with tables, classifications, or metadata. User queries are embedded in the same semantic space and matched against the knowledge base using cosine similarity, returning the most relevant items.

Building on this approach, the following section illustrates how semantic search has been implemented within IstatData, enabling natural-language queries and improving access to official statistical datasets.

V. IstatData: An AI-Enhanced Platform for Accessing Official Statistics

IstatData is Istat's data dissemination platform[1]. It offers thousands of multidimensional tables, interactive visualizations, and metadata describing statistical indicators. As part of Istat's modernization strategy, the platform has recently been enhanced with AI-based features designed to improve discoverability, accessibility, and user experience. Semantic search is central to this transformation, allowing users to interact with the system in natural language and retrieve relevant datasets more intuitively.

[1] IstatData platform is publicly available at the following link: https://esploradati.istat.it/databrowser/

VA. Semantic Search in IstatData

The semantic-search component follows the methodology described in the previous section. Table metadata—such as titles, descriptions, indicators, and analysis dimensions—are used to build the knowledge base. Each metadata element is transformed into a vector using a sentence-embedding model [9] and stored in the knowledge database. When a user submits a natural-language query, the system encodes the query with the same model, finds the most similar metadata entries using cosine similarity, and returns the related datasets (see

Figure 2).

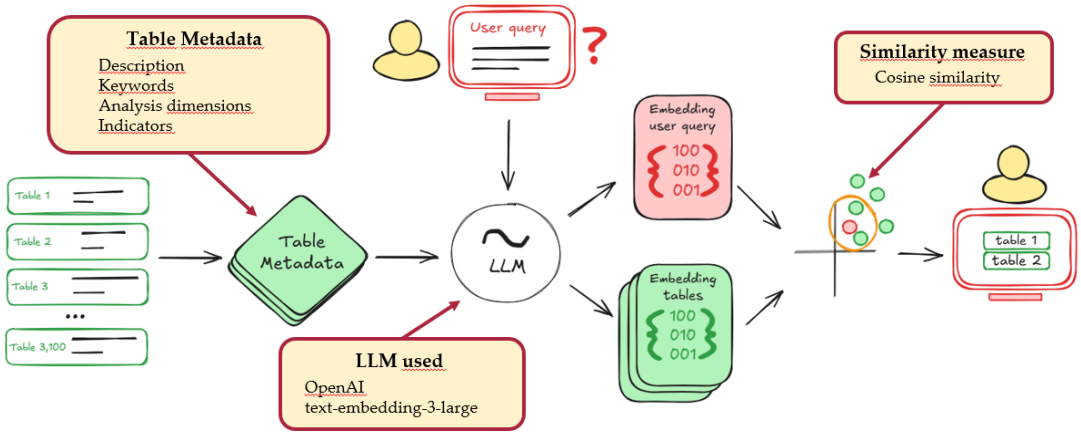

Figure 2: Semantic search workflow in IstatData. Table metadata (including titles, descriptions, keywords, and indicators) are converted into sentence embeddings and stored in a knowledge base. User queries are embedded in the same semantic space and used to ***query the knowledge base*** *through cosine similarity. The system then retrieves the datasets whose embeddings are closest to the user's query.*

This approach allows users to find statistical content even when they do not know the exact terminology used by official metadata. Queries such as *"economic competitiveness"*, *"enterprises exporting goods"*, or *"drinking habits of Italian population"* are successfully mapped to the appropriate datasets, improving accessibility for non-expert audiences and reducing navigation barriers in complex statistical domains.

The system was tested through relevance assessments, robustness evaluations, and safeguard procedures designed to identify model weaknesses, such as inappropriate responses or poor performance with noisy queries. These insights guided improvements in the embedding approach and metadata descriptors.

VB. Research Assistant: Retrieving Relevant Datasets

The **Research Assistant** feature allows users to search for data using natural language. When a query is submitted, the system retrieves all relevant datasets and presents them in a structured list. Users can view:

- dataset previews,
- associated tables, charts, and maps,
- metadata and definitions through the integrated glossary.

This feature supports users unfamiliar with statistical terminology and simplifies the initial steps of exploratory analysis. It also accelerates data retrieval for more experienced users who need rapid access to specific indicators. Figure 3 provides an example related to the user query "*give me data on economic competitiveness*".

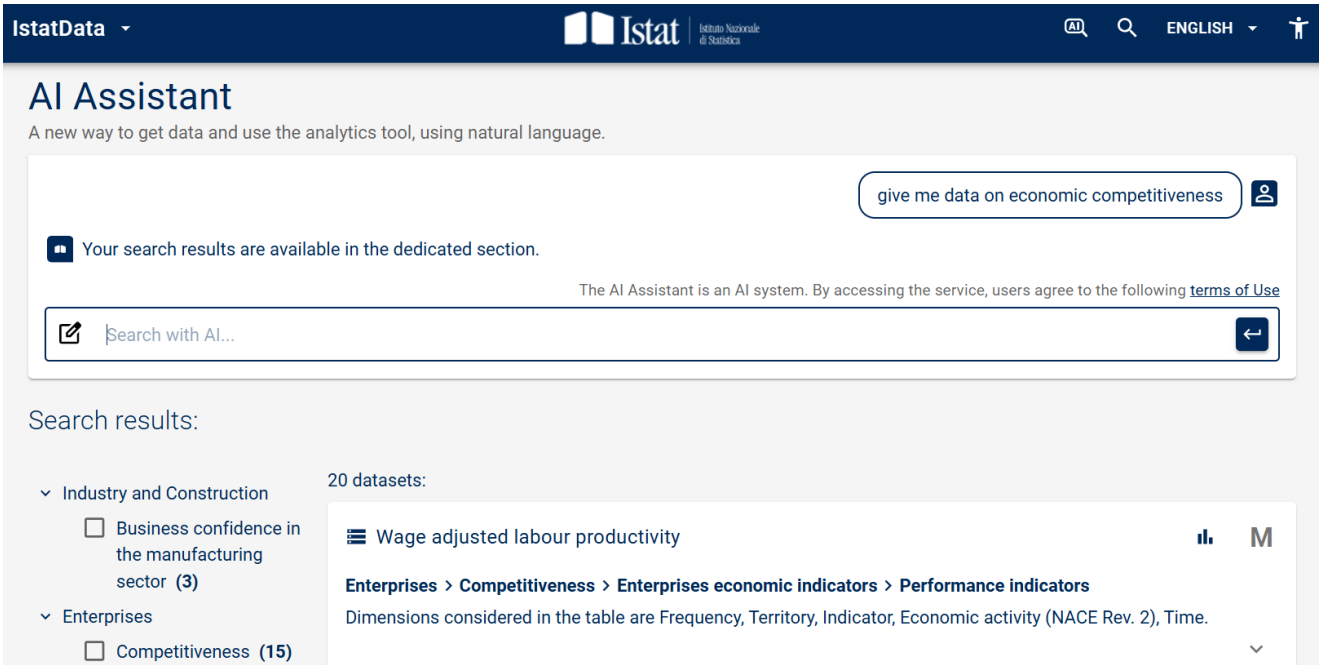

Figure 3: IstatData AI Assistant – Natural-language retrieval of statistical datasets. The AI Assistant allows users to submit queries in natural language and automatically retrieves the most relevant datasets. Search results are organised by domain, providing direct access to tables and metadata associated with the query.

VC. Support Assistant: Customising and Interpreting Data

A second component, the **Support Assistant**, provides AI-based tools to help users interact with statistical tables. Users can request operations such as:

- filtering or managing rows and columns,
- generating customised charts and maps,
- obtaining short textual descriptions of trends and indicators.

These functionalities simplify the navigation of large tables and enhance the interpretability of statistical information. They also support accessibility objectives by providing simplified narrative explanations for non-expert users.

Figure 4: IstatData AI Assistant – Support for chart exploration and interpretation. The AI Assistant guides users through statistical visualisations by allowing natural-language requests such as generating a chart or describing its content. The system provides contextual explanations, helping users interpret trends and indicators shown in the visualisation.

VI. Opportunities and Challenges of AI in Official Statistics

The experiments and applications presented in this paper highlight how AI can generate substantial value across the statistical domain—from data acquisition and processing to dissemination and user support. Modern AI techniques facilitate the integration of new data sources, improve the accessibility and granularity of statistical outputs, and enable more intuitive interactions between users and complex statistical systems. Platforms like IstatData demonstrate how semantic search, natural-language interfaces, and automated assistance can make official statistics easier to discover, customise, and interpret for a broad audience.

At the same time, adopting AI in official statistics requires careful evaluation of its associated risks. Black-box models may produce results that are difficult to interpret or validate against established quality standards [10]. Skill gaps and cultural resistance within organisations can slow down innovation, reducing the effectiveness of new tools. In addition, the increasing use of non-traditional and potentially sensitive data sources amplifies concerns about privacy, confidentiality, and data security. These risks underscore the need for robust governance frameworks, transparent methodological practices, and continuous investment in staff training and institutional capabilities. Several recent guidelines emphasise the importance of governance and institutional accountability when integrating AI into statistical systems [8,11].

Table 1 summarizes the main opportunities and challenges in adopting AI in official statistics.

In this context, the adoption of AI must remain aligned with the core values of official statistics—trust, transparency, and quality. Successful integration requires balancing innovation with responsibility, combining technological opportunities with the safeguards needed to ensure that AI-driven systems remain reliable, interpretable, and fully compliant with legal and ethical frameworks.

Table 1: Opportunities and challenges in adopting AI for official statistics

Opportunities	Challenges
AI adds value across the statistical workflow, supporting data acquisition, processing, dissemination, and user assistance.	**Black-box AI systems** may produce outputs that are difficult to interpret, validate, or trace back to source data.
New data sources and modern AI methods make official statistics more detailed, timely, and accessible to a broader range of users.	**Lack of skills and cultural resistance** can hinder the adoption of new tools and slow down innovation inside statistical organisations.
AI-enabled platforms such as IstatData improve discoverability, customisation, and understanding of statistical information.	**Privacy and data security risks, especially** when analysing non-traditional or sensitive data, require robust safeguards and legal compliance.
AI-driven automation can reduce manual burden and increase the efficiency of coding, editing, and classification tasks.	**Infrastructure constraints** (e.g., limited GPU capacity, legacy systems) may limit the scalability of AI solutions.
Multidisciplinary collaboration between methodologists, data scientists, and IT experts strengthens innovation capacity.	**Governance demands** are higher: AI requires clear responsibilities, validation procedures, and transparent methodological documentation.

VII. Conclusion

The experience presented in this paper shows that artificial intelligence can meaningfully support the modernisation of official statistics, both by enabling the use of complex, non-traditional data sources and by improving the accessibility and usability of statistical products. The integration of semantic search, natural-language interfaces, and automated assistance within IstatData illustrates how AI can help users navigate large statistical repositories and extract value from official information more intuitively.

At the same time, the adoption of AI in National Statistical Institutes must continue to be guided by the principles that support trust in official statistics: methodological soundness, transparency, privacy protection, and institutional accountability. Ongoing work on governance, validation, and staff training is therefore essential to ensure that AI-driven solutions stay reliable, interpretable, and aligned with legal and ethical standards.

Looking forward, further developments will focus on expanding semantic capabilities, strengthening robustness and quality assurance procedures, and exploring multimodal approaches that integrate text, tables, charts, and maps. Continued collaboration within the European Statistical System and the wider international community will play a crucial role in shaping responsible and sustainable uses of AI for official statistics.

Ultimately, AI should not replace the statistical expertise that characterizes National Statistical Institutes, but rather enhance it—supporting new ways of producing and sharing data while preserving the quality, trust, and public value that official statistics are meant to provide.

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